

FINNY NOVEL BALAGAM

Software Engineer | Backend Developer | Distributed Systems
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PROFESSIONAL SUMMARY

High-impact Software Engineer with expertise in building fault-tolerant backend architectures, low-latency RESTful microservices, and high-throughput real-time data pipelines. Proven proficiency in Java 17, Spring Boot 3.x, Python, and SQL, complemented by hands-on experience with Apache Kafka, Spark, Docker, and Kubernetes. Adept at applying SOLID principles and OOP design patterns to decouple complex software systems, optimize database query latency, and automate robust CI/CD deployment pipelines.

TECHNICAL SKILLS

Languages:	Java (Java 17), Python, JavaScript (ES6+), SQL
Backend & Frameworks:	Spring Boot 3.x, Spring Data JPA, Node.js, Express.js, Flask, REST APIs, JWT Authentication
Distributed Systems:	Apache Kafka, Apache Spark (Structured Streaming)
Databases & DevOps:	PostgreSQL, MongoDB, Docker, Kubernetes, GitHub Actions, Maven, Git, Linux, AWS, Render
Core Concepts:	OOP, SOLID Principles, Design Patterns (Factory, Adapter, Observer, Singleton), DBMS, Operating Systems, Computer Networks, System Design

PROJECTS

Enterprise Multi-Platform Notification Engine

Java 17, Spring Boot 3.x, PostgreSQL, Spring Data JPA, Docker, Kubernetes, GitHub Actions, Maven, REST APIs

- Engineered a high-throughput backend notification microservice strictly adhering to SOLID principles and design patterns (Singleton, Factory, Adapter, Observer).
- Decoupled third-party vendor APIs (Twilio, SendGrid) using structural Adapters, safeguarding core logic against external breaking changes and network timeouts.
- Implemented dynamic Factory Pattern to instantiate communication channels (SMS, EMAIL) at runtime based on payload configurations for seamless protocol extensibility.
- Architected asynchronous Observer pattern to handle post-delivery audit logging and metrics tracking without blocking primary request execution.
- Containerized microservices using multi-stage Docker builds, orchestrated deployments via Kubernetes, and built automated GitHub Actions CI/CD pipelines.

TrafficAdvisor | Real-Time Traffic Monitoring System

Python · Apache Kafka · Apache Spark (Structured Streaming) · PostgreSQL/PostGIS · Flask · Leaflet.js

- Architected an end-to-end real-time data pipeline, building a custom Python Kafka producer to stream traffic sensor events into a Spark Structured Streaming job for downstream processing.
- Built a Spark Structured Streaming job to parse, transform, and persist Kafka event streams to PostgreSQL in near real time via micro-batch writes over JDBC.
- Implemented PostGIS with GIST spatial indexing to power fast, geo-indexed nearest-neighbor lookups for the "nearest sensor" query.
- Developed a Flask REST API exposing live traffic metrics through geospatial queries, powering an interactive Leaflet/OpenStreetMap interface for real-time, click-to-query lookups.

Hotel Room Reservation System

Node.js, Express, React, MongoDB, JWT, Render, Netlify

- Architected and deployed a full-stack booking platform as three independently hosted services (React client, React admin dashboard, Express API) across Netlify, Render, and MongoDB Atlas, configuring CORS and environment-based API routing for production.
- Designed and built 20+ RESTful API endpoints across user, hotel, and room resources, secured with JWT authentication (httpOnly cookies) and role-based route middleware separating public, authenticated, and admin-only access.
- Modeled room availability at the database level, tracking reserved date ranges per room number and disabling already-booked dates in the booking calendar to prevent guests from selecting them.
- Resolved production deployment issues including cross-origin request failures, auth state persistence, and route conflicts across the three separately hosted apps.

EXPERIENCE

APSCHE SkillForge

06/2025 - 08/2025

Web Development Intern

- Engineered scalable backend service logic and RESTful API endpoints using Node.js and Express with MongoDB persistence.
- Optimized complex database queries to significantly reduce query latency, improve throughput, and elevate system response speed.
- Executed comprehensive API endpoint testing using structured JSON payloads to ensure high reliability and edge-case coverage.

EDUCATION

Velagapudi Ramakrishna Siddhartha Engineering College

Graduation: 2026

B.Tech in Artificial Intelligence & Data Science | CGPA: 8.2/10

CERTIFICATIONS & ACHIEVEMENTS

- ServiceNow Certifications:** Certified System Administrator (CSA) & Certified Application Developer (CAD).
- Cisco & Linux Certifications:** Python Programming Essentials & NDG Linux Essentials Certification.
- Competitive Programming:** Solved 400+ algorithmic data structure problems on LeetCode & HackerRank.
- Honors:** Awarded prestigious NSF Scholarship for stellar academic excellence and technical achievements.